Quy-Sang Mai, CFA

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Research interests: ML-driven signal combination for abnormal returns in emerging equity markets; investor sentiment and earnings-surprise prediction.

EDUCATION

MSc, Applied Mathematics (Quantitative & Computational Finance) *2016 - 2019*
University of Science, VNU-HCM - Capstone graded 10/10; Stochastic Calculus 9.5/10, Probability & Statistics 9.0/10

Bachelor of Business (Economics and Finance) *2012 - 2015*
RMIT University - Awarded with Distinction

Mathematics (specialised), High School for the Gifted, VNU-HCMC *2009 - 2012*

RESEARCH EXPERIENCE

Capstone Project (graded 10/10) - University of Science, VNU-HCM *Jan - Mar 2018*
Built a three-layer system for VN30 equities - gradient-boosted decision-tree 'experts' over ~40 technical signals, combined by an online expert-weighting layer (**Creamer & Freund 2010**) with a risk overlay - to predict the sign of next-day abnormal returns vs the index. Out-of-sample (2015-17) it beat the VN30 on risk-adjusted measures (Information / Sharpe ratios ~2-4), with a transaction-cost sensitivity analysis confirming outperformance persisted to ~2 bps per trade. The same signal-combination approach is what the proposed PhD would extend to text- and LLM-derived sentiment signals for earnings-surprise prediction. Supervised by **Dr Minh Dang** (State Street, Hong Kong).

Master's Thesis - University of Science, VNU-HCM *2019*
"Smile-adjusted delta hedging for options improved with boosting." Added a gradient-boosted-regression-tree step to the minimum-variance delta of **Hull and White (2017)** ; on ~1.3M S&P 500 index option records (OptionMetrics, 2015-17) it delivered a statistically significant out-of-sample reduction in hedging error (Newey-West t-tests) for put options. Supervised by **Dr Minh Dang** .

Research Assistant - RMIT University Vietnam *Aug 2016 - Mar 2017*
Co-authored learning-and-teaching research with **A/Prof. Mathews Nkhoma** : literature review, database searches, and qualitative & quantitative data analysis, resulting in the peer-reviewed conference paper listed below.

PUBLICATIONS

Nkhoma, C. A., Nkhoma, M., Ulhaq, I., & **Mai, S. Q.** (2017). Enhancing students' learning through early class preparation. Proceedings of InSITE 2017, 85-95. doi:10.28945/3756

TEACHING

Teaching Assistant - "Derivatives Pricing in Practice" (graduate course) *2017 - 2018*
University of Science, VNU-HCM. Supported delivery of the graduate derivatives-pricing course under **Dr Minh Dang** .

PROFESSIONAL EXPERIENCE

Ho Chi Minh City Securities Corporation (HSC) *2017 - Present*
One of Vietnam's leading brokerage and investment-banking firms.

Manager / Senior Manager - Structured Products & Quantitative Trading *Jan 2024 - Present*
- Lead quantitative research - developing and backtesting models to construct equity portfolios optimised for absolute and risk-adjusted returns;
- Oversee the structured-products (covered warrants) desk - pricing, market-making, hedging and reporting;
- Mandated to build out and lead structured-products trading (equity options and other new structured products) as they launch in the Vietnamese market.

Senior / Covered Warrant Trader; ETF Trader *Sep 2017 - Dec 2023*
- Product development, pricing, market-making and reporting for HSC's covered-warrant and ETF portfolios (authorised participant); backup coverage for index arbitrage.

Investment Analyst - Saigon Asset Management (SAM) *Aug 2014 - Aug 2015*
Research and valuation of Vietnamese listed companies (DCF and comparables) for funds with peak AUM over USD$200m.

TECHNICAL SKILLS

Econometrics & event studies - Supervised ML (gradient boosting, online learning) - Out-of-sample & after-cost backtesting - Python (pandas, scikit-learn) - C++ - LaTeX

CREDENTIALS

CFA Charterholder (2022) - Fund Management Professional License, Vietnam (2023)

REFEREES

Available on request.